

Linear Independence

1.7 Linear Independence

Homogeneous Vector Equations

The homogeneous equations in Section 1.5 can be studied from a different perspective by writing them as vector equations. In this way, the focus shifts from the unknown solutions of $A\mathbf{x} = \mathbf{0}$ to the vectors that appear in the vector equations.

For instance, consider

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (1)$$

This equation has the trivial solution $x_1 = x_2 = x_3 = 0$. The main issue is whether the trivial solution is the only solution.

Linear Independence

Definition

An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution. The set is **linearly dependent** if there exist weights $c_1, \dots, c_p$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}. \quad (2)$$

Equation (2) is called a **linear dependence relation** among the vectors when the weights are not all zero.

Example 1

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

- a. Determine whether $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.
- b. If possible, find a linear dependence relation among the three vectors.

Solution idea

We must determine if there is a nontrivial solution of the equation on the previous slide.

Example 1: Row Reduction

Row operations on the associated augmented matrix show that

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

- ▶ x_1 and x_2 are basic variables, and x_3 is free.
- ▶ Each nonzero value of x_3 determines a nontrivial solution of (1).
- ▶ Hence $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent.

Example 1: A Dependence Relation

To find a linear dependence relation among $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, completely row reducing the augmented matrix gives

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \begin{array}{l} x_1 - 2x_3 = 0, \\ x_2 + x_3 = 0. \\ 0 = 0. \end{array}$$

- ▶ Thus $x_1 = 2x_3$, $x_2 = -x_3$, and x_3 is free.
- ▶ Choose, for example, $x_3 = 5$.
- ▶ Then $x_1 = 10$ and $x_2 = -5$.

Example 1: Conclusion

- ▶ Substituting the values into the vector equation gives

$$10\mathbf{v}_1 - 5\mathbf{v}_2 + 5\mathbf{v}_3 = \mathbf{0}.$$

- ▶ This is one of infinitely many possible linear dependence relations among $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

Columns of a Matrix

Suppose we begin with a matrix $A = [\mathbf{a}_1 \cdots \mathbf{a}_n]$ instead of a set of vectors. Then the matrix equation

$$A\mathbf{x} = \mathbf{0}$$

can be written as

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}.$$

- ▶ *Each linear dependence relation among the columns of A corresponds to a nontrivial solution of $A\mathbf{x} = \mathbf{0}$.*
- ▶ The columns of A are linearly independent if and only if $A\mathbf{x} = \mathbf{0}$ has *only the trivial solution*.

Sets of One Vector

- ▶ A set containing only one vector, say $\mathbf{v}$, is linearly independent if and only if $\mathbf{v}$ is not the zero vector $\mathbf{0}$.
- ▶ This is because the vector equation $x_1\mathbf{v} = \mathbf{0}$ has only the trivial solution when $\mathbf{v} \neq \mathbf{0}$.
- ▶ The zero vector is linearly dependent because $x_1\mathbf{0} = \mathbf{0}$ has many nontrivial solutions.

Sets of Two Vectors

- ▶ A set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent if and only if neither vector is a multiple of the other.
- ▶ The set is linearly dependent if at least one vector is a scalar multiple of the other.

Sets of Two or More Vectors

THEOREM 7

Characterization of Linearly Dependent Sets. An indexed set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one vector in S is a linear combination of the others. In fact, if S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ with $j > 1$ is a linear combination of the preceding vectors $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$.

Using Theorem 7

- ▶ Theorem 7 *does not* say that *every* vector in a linearly dependent set is a linear combination of the preceding vectors.
- ▶ A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

Example 4. Let

$$\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}.$$

Describe $\text{Span}\{\mathbf{u}, \mathbf{v}\}$, and explain why $\mathbf{w}$ is in this span if and only if $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent.

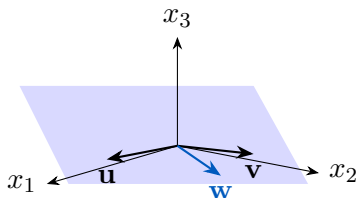
Example 4: Solution

- **Solution:** $\mathbf{u}$ and $\mathbf{v}$ are linearly independent because neither is a multiple of the other; consequently, they span a plane in

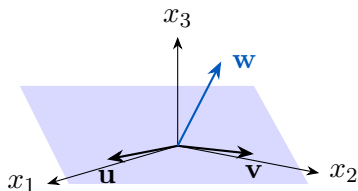
Example 4: Solution

- ▶ **Solution:** $\mathbf{u}$ and $\mathbf{v}$ are linearly independent because neither is a multiple of the other; consequently, they span a plane in $\mathbb{R}^3$.
- ▶ $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is the x_1x_2 -plane, where $x_3 = 0$.
- ▶ If $\mathbf{w}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$, then $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent by Theorem 7.
- ▶ Conversely, if $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is dependent
- ▶ By Theorem 7, some vector in $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a linear combination of the preceding vectors. Since $\mathbf{u} \neq 0$ and $\mathbf{u}, \mathbf{v}$ are independent, that vector must be $\mathbf{w}$.

Example 4: Geometric View



Linearly dependent: $\mathbf{w} \in \text{Span}\{\mathbf{u}, \mathbf{v}\}$.



Linearly independent: $\mathbf{w} \notin \text{Span}\{\mathbf{u}, \mathbf{v}\}$.

- The example 4 generalizes to any set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ in $\mathbb{R}^3$ with $\mathbf{u}$ and $\mathbf{v}$ linearly independent.
- The set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ will be linearly dependent if and only if $\mathbf{w}$ is in the plane spanned by $\mathbf{u}, \mathbf{v}$.

A Set Containing the Zero Vector

THEOREM 9

If a set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ contains the zero vector, then the set is linearly dependent.

Proof. Renumber the vectors so that $\mathbf{v}_1 = \mathbf{0}$. Then

$$1\mathbf{v}_1 + 0\mathbf{v}_2 + \dots + 0\mathbf{v}_p = \mathbf{0}$$

is a linear dependence relation because the weights are not all zero.

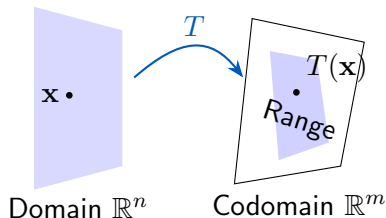
► LINEAR TRANSFORMATIONS

(Section 1.8)

Transformations

A **transformation** (function or mapping) T from $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x} \in \mathbb{R}^n$ and a vector $T(\mathbf{x}) \in \mathbb{R}^m$.

- ▶ The set $\mathbb{R}^n$ is the **domain** of T , and the set $\mathbb{R}^m$ is the **codomain** of T .
- ▶ The notation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$.
- ▶ For $\mathbf{x} \in \mathbb{R}^n$, the vector $T(\mathbf{x}) \in \mathbb{R}^m$ is called the **image of $\mathbf{x}$ (under the action of T)**.



Domain, codomain, and range of T :

$$\mathbb{R}^n \rightarrow \mathbb{R}^m$$

Matrix Transformations

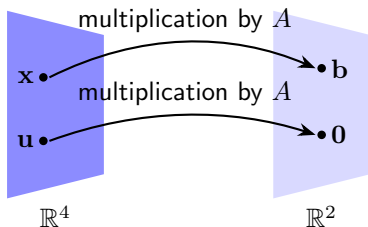
- ▶ For each $\mathbf{x} \in \mathbb{R}^n$, $T(\mathbf{x})$ is computed as $A\mathbf{x}$, where A is an $m \times n$ matrix.
- ▶ For simplicity, we denote such a *matrix transformation* by $x \mapsto Ax$
- ▶ Observe that the domain of T is $\mathbb{R}^n$ when A has n columns and the codomain of T is $\mathbb{R}^m$ when each column of A has m entries.
- ▶ The range of T is the set of all linear combinations of the columns of A , because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$.

Transforming Vectors by Matrix Multiplication

For instance, the equations

$$\underbrace{\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 5 \\ 8 \end{bmatrix}}_{\mathbf{b}}, \quad A \underbrace{\begin{bmatrix} 4 \\ -1 \\ 3 \\ -3 \end{bmatrix}}_{\mathbf{u}} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{\mathbf{0}}.$$

say that multiplication by A transforms $\mathbf{x}$ into $\mathbf{b}$ and transforms $\mathbf{u}$ into the zero vector. See Figure below.



Matrix Transformations: Example 1

Let

$$A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}.$$

and define a transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$. Thus

$$T(\mathbf{x}) = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}.$$

Matrix Transformations: Questions

- a. Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .
- b. Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$.
- c. Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?
- d. Determine whether $\mathbf{c}$ is in the range of the transformation T .

Matrix Transformations: Solution

Solution:

- a. Compute

$$T(\mathbf{u}) = A\mathbf{u} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}.$$

- b. Solve $T(\mathbf{x}) = \mathbf{b}$ for $\mathbf{x}$. That is, solve $Ax = b$, or

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}. \quad (1)$$

Matrix Transformations: Row Reduction

Row reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 1.5 \\ 0 & 1 & -.5 \\ 0 & 0 & 0 \end{array} \right] \quad (2)$$

Hence $x_1 = 1.5$, $x_2 = -.5$, and

$$\mathbf{x} = \begin{bmatrix} 1.5 \\ -.5 \end{bmatrix}.$$

The image of this $\mathbf{x}$ under T is the given vector $\mathbf{b}$.

Matrix Transformations: Uniqueness and Range

- c. Any $\mathbf{x}$ whose image under T is $\mathbf{b}$ must satisfy equation (1).
 - ▶ From (2), it is clear that equation (1) has a unique solution.
 - ▶ So there is exactly one $\mathbf{x}$ whose image is $\mathbf{b}$.
- d. The vector $\mathbf{c}$ is in the range of T if $\mathbf{c}$ is the image of some $\mathbf{x}$ in $\mathbb{R}^2$, that is, if $\mathbf{c} = T(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^2$.

This is another way of asking if the system

$$A\mathbf{x} = \mathbf{c}$$

is consistent.

Matrix Transformations: Is $\mathbf{c}$ in the Range?

To find the answer, row reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{array} \right].$$

The third equation, $0 = -35$, shows that the system is inconsistent.

So $\mathbf{c}$ is not in the range of T .

Four Basic Questions About a Transformation

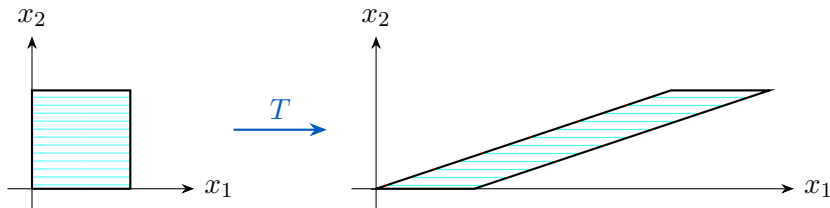
For $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$:

- ❶ Question 1: for any $\mathbf{u} \in \mathbb{R}^n$, what is the image of $T(\mathbf{u})$? (function value)
- ❷ Question 2: find an $\mathbf{x} \in \mathbb{R}^n$ whose image under T is $\mathbf{b}$. (independent value)
- ❸ Question 3: is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?
- ❹ Determine if $\mathbf{c}$ is in the range of the transformation T .

Shear Transformation (剪切变换)

Example 3 Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. The transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\mathbf{x}) = A\mathbf{x}$ is called a **shear transformation**.

- It can be shown that if T acts on each point in the 2×2 square shown below, then the set of images forms the shaded parallelogram.



Shear Transformation

The key idea is to show that T maps line segments onto line segments, and then to check that the corners of the square map onto the vertices of the parallelogram.

For instance, the image of the point $u = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

$$T(u) = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix},$$

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}.$$

T deforms the square as if its top were pushed right while its base stayed fixed.

Linear Transformations

Definition

A transformation (or mapping) T is **linear** if

- ① $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in its domain;
- ② $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in its domain.

Every matrix transformation is linear. Important examples of linear transformations that are not matrix transformations will be discussed in Chapters 4 and 5.

Linear Transformations Preserve Operations

Linear transformations *preserve the operations of vector addition and scalar multiplication*.

- ▶ $T(\mathbf{u} + \mathbf{v})$ can be found either by adding first $\mathbf{u}$ and $\mathbf{v}$ and then applying T to $\mathbf{u} + \mathbf{v}$, or by applying T first towards $T(\mathbf{u})$ and $T(\mathbf{v})$, and adding the images to $T(\mathbf{u} + \mathbf{v})$.
- ▶ These two properties lead to the following useful facts

$$T(\mathbf{0}) = \mathbf{0}, \quad (3)$$

and, for all scalars c, d ,

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v}). \quad (4)$$

Why Properties (3) and (4) Hold

Property (3) follows from condition (2) in the definition because:

$$T(\mathbf{0}) = T(0\mathbf{u}) = 0T(\mathbf{u}) = \mathbf{0}.$$

Property (4) requires both defining properties:

$$T(c\mathbf{u} + d\mathbf{v}) = T(c\mathbf{u}) + T(d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v}).$$

A useful converse

If a transformation satisfies (4) for all $\mathbf{u}, \mathbf{v}$ and c, d , it must be linear.

(Set $c = d = 1$ for preservation of addition; set $d = 0$ for preservation of scalar multiplications.)

The Superposition Principle

Repeated application of (4) gives

$$T(c_1 \mathbf{v}_1 + \cdots + c_p \mathbf{v}_p) = c_1 T(\mathbf{v}_1) + \cdots + c_p T(\mathbf{v}_p). \quad (5)$$

In engineering and physics, (5) is called the *superposition principle* (疊加原理) .

Contractions and Dilations

Given a scalar r , define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T(\mathbf{x}) = r\mathbf{x}.$$

Contraction

$0 \leq r \leq 1$: vectors keep their direction and shrink toward the origin.

Dilation

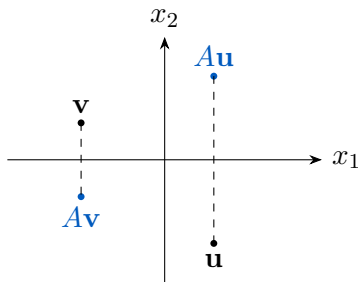
$r > 1$: vectors keep their direction and stretch away from the origin.

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From a Matrix to a Geometric Description

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Given a geometric description of the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

Plot some random points (vectors) on graph paper to see what happens. A point such as $(4,1)$ maps into $(4,-1)$. The transformation $\mathbf{x} \mapsto A\mathbf{x}$ reflects points through the x -axis, or x_1 -axis.



A question

Given a geometric description of a linear transformation,

how do we describe the transformation?

(How to give an explicit function relationship between $\mathbf{x}$ and $\mathbf{y}$)

A “formula” for $T(\mathbf{x})$.

Transformation and Matrix

Every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$



a matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$



the matrix A

The Matrix of a Linear Transformation (Section 1.9)

Theorem 10. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n. \quad (3)$$

In fact, A is the $m \times n$ matrix whose j th column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j th column of the identity matrix in $\mathbb{R}^n$:

The Matrix of a Linear Transformation: Proof

Proof: Write

$$\mathbf{x} = I_n \mathbf{x} = [\mathbf{e}_1 \ \cdots \ \mathbf{e}_n] \mathbf{x} = x_1 \mathbf{e}_1 + \cdots + x_n \mathbf{e}_n.$$

and use the linearity of T to compute

$$T(x) = T(x_1 e_1 + \cdots + x_n e_n) = x_1 T(e_1) + \cdots + x_n T(e_n)$$

The Standard Matrix

The matrix A in (3) is called the **standard matrix for the linear transformation T** .

We know now that every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be viewed as a matrix transformation, and vice versa.

- ▶ The term *Linear transformation* focuses on a property of a mapping.
- ▶ The term *Matrix transformation* describes how such a mapping is implemented, as the example on the next slide illustrates.

Example 2: Standard Matrix of a Dilation

Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$ in $\mathbb{R}^2$.

Solution: Write

$$T(\mathbf{e}_1) = 3\mathbf{e}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = 3\mathbf{e}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

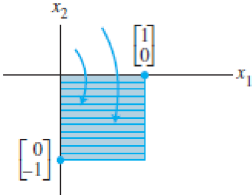
Therefore

$$A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2)] = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

Geometric Linear Transformations of $\mathbb{R}^2$

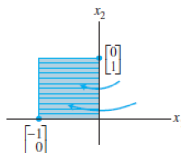
Tables 1–4 illustrate common geometric linear transformations of the plane.

TABLE 1 Reflections

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

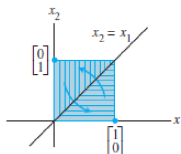
Reflections: Table 1 Continued

Reflection through
the x_2 -axis



$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

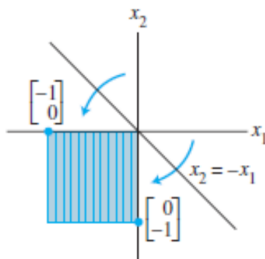
Reflection through
the line $x_2 = x_1$



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

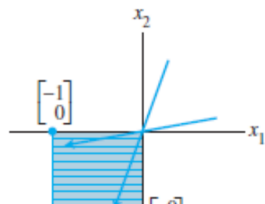
Reflections: Table 1 Continued

Reflection through
the line $x_2 = -x_1$



$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Reflection through
the origin



$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Table 2: Contractions and Expansions

TABLE 2 Contractions and Expansions

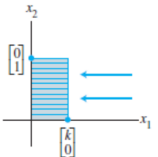
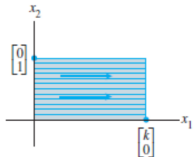
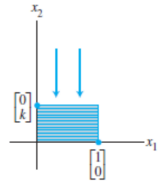
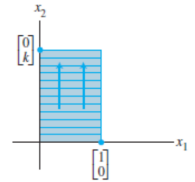
Transformation	Image of the Unit Square		Standard Matrix
Horizontal contraction and expansion			$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$
	$0 < k < 1$	$k > 1$	
Vertical contraction and expansion			$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$
	$0 < k < 1$	$k > 1$	

Table 3: Shears

TABLE 3 Shears

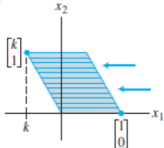
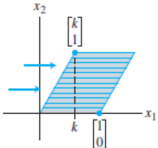
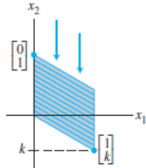
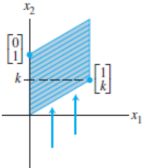
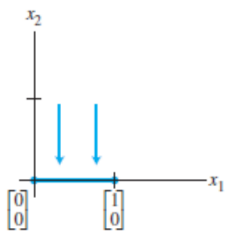
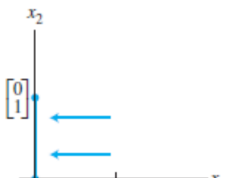
Transformation	Image of the Unit Square	Standard Matrix
Horizontal shear	 $k < 0$  $k > 0$	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
Vertical shear	 $k < 0$  $k > 0$	$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$

Table 4: Projections

TABLE 4 Projections

Transformation	Image of the Unit Square	Standard Matrix
Projection onto the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
Projection onto the x_2 -axis		$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

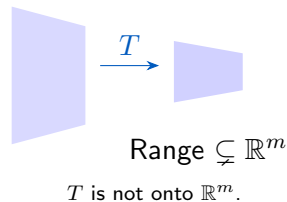
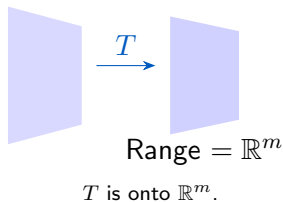
Existence and Uniqueness Questions

Definition

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if each $\mathbf{b} \in \mathbb{R}^m$ is the image of at least one $\mathbf{x} \in \mathbb{R}^n$.

- ▶ Equivalently, T is onto $\mathbb{R}^m$ when the range of T is all of the codomain $\mathbb{R}^m$.
- ▶ That is, T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if, for each $\mathbf{b}$ in the codomain $\mathbb{R}^m$, there exists at least one solution of $T(\mathbf{x})=\mathbf{b}$.
- ▶ "Does T map $\mathbb{R}^n$ onto $\mathbb{R}^m$?" is an existence question.
- ▶ The mapping T is not onto $\mathbb{R}^m$ when there is some $\mathbf{b}$ in $\mathbb{R}^m$ for which the equation $T(\mathbf{x})=\mathbf{b}$ has no solution. See the figure on the next slide.

Onto and One-to-One



Definition of One-to-one

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be **one-to-one** if every $\mathbf{b} \in \mathbb{R}^m$ is the image of *at most one* $\mathbf{x} \in \mathbb{R}^n$.

Example 4: Onto and One-to-One

Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$? Is T a one-to-one mapping?

Example 4: Solution

Solution:

- ▶ Since A is in echelon form, we can see at once that A has a pivot position in each row.
- ▶ By Theorem 4 in Section 1.4, for each $\mathbf{b}$ in $\mathbb{R}^3$, the equation $A\mathbf{x}=\mathbf{b}$ is consistent.
- ▶ In other words, the linear transformation T maps $\mathbb{R}^4$ (its domain) onto $\mathbb{R}^3$.

However, since $A\mathbf{x} = \mathbf{b}$ has a **free variable** because there are four variables and only three basic variables, each $\mathbf{b}$ is the image of more than one $\mathbf{x}$. Thus T is not one-to-one.

Example 4: recap

- ▶ Check the echelon form of linear transformation:
- ▶ If consistent: maps domain to codomain.
- ▶ If and only if no free variable: one-to-one mapping

Kernel and Range of a Linear Transformation

Subspaces of vector spaces other than $\mathbb{R}^n$ are often described in terms of a linear transformations instead of matrices.

Definition

A **linear transformation** T from a vector space V into a vector space W is a rule that assigns to each $\mathbf{x} \in V$ a unique vector $T(\mathbf{x}) \in W$, such that

- ❶ $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$;
- ❷ $T(c\mathbf{u}) = cT(\mathbf{u})$ for every $\mathbf{u} \in V$ and scalar c .

The **kernel** is $\{\mathbf{x} \in V : T(\mathbf{x}) = \mathbf{0}\}$; the **range** is $\{T(\mathbf{x}) : \mathbf{x} \in V\}$.

Glossary

- | | |
|----------------------------------|---------|
| 1. Square matrix | 方阵 |
| 2. (Row) Column Vector | (行)列向量 |
| 3. Parallelogram rule | 平行四边形法则 |
| 4. Scalar multiplication | 数乘 |
| 5. Linear combination | 线性组合 |
| 6. Parametric vector form | 参数向量形式 |
| 7. (Non)Homogeneous | (非)齐次的 |
| 8. (Non)Trivial solution | (非)平凡解 |